Climate-Conscious & Cost-Effective: The Case for Plant-Based University Catering

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Executive Summary

Climate change poses a serious threat to many aspects of human society. Secretary General of the UN Antonio Guterres has called for "climate action on all fronts – everything, everywhere, all at once"[1]. Our food systems - responsible for approximately one-third of all greenhouse gas (GHG) emissions - have a significant impact on the climate crisis. Of this, the impact of animal agriculture is greatly outsized, whereas that of plant-based foods is comparatively much lower. There is a clear and urgent need for a just and sustainable transition towards plant-based institutional catering services. Universities, as the institutions that have made clear the threat of rapid climate change and as stewards of future generations, have a well-placed responsibility to lead this transition.

This report models the financial and environmental impacts of a transition to plant-based catering in universities. We compiled environmental impact data, wholesale price, and portion sizes for 140 different ingredients to estimate the environmental footprint and food procurement costs for a range of meat-based, vegetarian, and plant-based university meals.

Our major findings are:

1. A plant-based transition can be financially beneficial for universities. Indeed, **the transition presents an opportunity for cost savings, with food costs for plant-based meals being on average 30% cheaper than meat-based meals**, and 21% cheaper than vegetarian meals.

2. Plant-based meals greatly diminish the environmental impact of university catering. **Plant-based meals emit 84% less CO2eq on average compared to meat-based meals, and use less than one-tenth of the land**.

3. **A medium-sized university with 10,000 students could save over £500,000 (€600,000/$650,000) annually on food procurement costs** by adopting 100% plant-based university catering.

[1] United Nations. (2023, March 20). Intergovernmental panel report press release. Available at https://press.un.org/en/2023/sgsm21730.doc.htm

1. Background

Growing scientific consensus continues to emphasise the pressing need for a system-level shift away from animal product consumption and towards plant-based diets. The reasons for doing so are multiple and urgent.

Animal agriculture is a significant and neglected element of the climate crisis. Food systems contribute approximately 34% of all GHG emissions, with the livestock sector contributing an outsized proportion of these emissions - between 71%-80%[2]. This means that animal agriculture is responsible for approximately 24%-27% of all GHG emissions. Animal product production also demands significant land use, and is a leading cause of deforestation: in total, 80% of all agricultural land is used for meat and dairy production[3]. In contrast, plant-based foods have a significantly smaller environmental footprint[4,5,6].

As a result, the transition away from animal farming and towards alternatives is one of the most impactful and neglected climate change mitigation strategies available[7].

Public health issues including pandemic risk, antibiotic resistance, and lifestyle diseases are also associated with the consumption of animals. Certain animal products – particularly red and processed meat – are associated with an increased risk of common and burdensome chronic diseases including diabetes and cardiovascular disease,

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[2] Feigin, S. V., Wiebers, D. O., Lueddeke, G., Morand, S., Lee, K., Knight, A., ... & Winkler, A. S. (2023). Proposed solutions to anthropogenic climate change: A systematic literature review and a new way forward. *Heliyon*.
[3] Our World In Data. (2021, March 4). If the world adopted a plant-based diet, we would reduce global agricultural land use from 4 to 1 billion hectares. Available at https://ourworldindata.org/land-use-diets
[4] Clark, M. A., Springmann, M., Hill, J., & Tilman, D. (2019). Multiple health and environmental impacts of foods. *Proceedings of the National Academy of Sciences*, 116(46).
[5] Errickson, F., Kuruc, K., & McFadden, J. (2021). Animal-based foods have high social and climate costs. *Nature Food*, 2(4), 274-281.
[6] Poore, J., & Nemecek, T. (2018). Reducing food's environmental impacts through producers and consumers. *Science*, 360(6392), 987-992.
[7] Feigin et al. (20230)

as well as certain cancers[8,9,10]. Processed meat is now classified by the WHO as a Group 1 carcinogen, in the same class as tobacco[11].

Transitioning towards plant-based public catering systems can accelerate the adoption of plant-based diets and mitigate the harms associated with farming animals. Plant-rich catering systems shift the burden of change away from the individual, and instead create environments where the most beneficial and sustainable options are most accessible.

Even very simple changes can have significant positive effects. A study by Sodexo – one of the world's largest foodservice providers – found that simply implementing plant-based options as the default meal in university canteens led to a 51 percentage point increase in students choosing a plant-based meal[12].. Moreover, students were also significantly more likely to express satisfaction with plant-based meals on days when they were the default. This suggests that these simple changes also serve to improve student satisfaction with plant-based meals and normalise plant-based eating beyond university canteens.

Other institutions are already successfully transitioning to plant-based catering systems. New York City Health + Hospitals, the largest municipal healthcare delivery system in the United States, serves default plant-based meals across New York City's 11 public hospitals. In 2024, it claimed to have decreased food-related emissions by 36%, recorded cost savings of around 59 cents per meal, and achieved a patient satisfaction rate above 90%.[13]

Both in research and in practice, plant-based public catering is proving to be a low-cost, high-impact means of reducing food-related emissions (among other benefits), without

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[8] Bouvard, V., Loomis, D., Guyton, K. Z., Grosse, Y., El Ghissassi, F., Benbrahim-Tallaa, L., ... & Straif, K. (2015). Carcinogenicity of consumption of red and processed meat. *The Lancet Oncology*, 16(16), 1599-1600.
[9] Godfray, H. C. J., Aveyard, P., Garnett, T., Hall, J. W., Key, T. J., Lorimer, J., ... & Jebb, S. A. (2018). Meat consumption, health, and the environment. *Science*, 361(6399).
[10] Musicus, A. A., Wang, D. D., Janiszewski, M., Eshel, G., Blondin, S. A., Willett, W., & Stampfer, M. J. (2022). Health and environmental impacts of plant-rich dietary patterns: a US prospective cohort study. *The Lancet Planetary Health*, 6(11).
[11] World Health Organisation. (2015, October 26). Cancer: Carcinogenicity of the consumption of red meat and processed meat. Available at https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat
[12] Food for Climate League (2023, May). Serving Up Plants By Default. Available at https://betterfoodfoundation.org/wp-content/uploads/2023/05/Exec-Summary_Serving-Up-Plants-by-Default.pdf
[13] Morgenstern, S., Redwood, M., & Herby, A. (2024). An Innovative Program for Hospital Nutrition. *American Journal of Lifestyle Medicine*.

raising catering costs or compromising on consumer satisfaction. Indeed, plant-based public catering can result in *reduced* catering costs and *increased* consumer satisfaction.

Hence, there is an urgent need for a transition towards plant-based university catering systems. Any transition must occur in stages, and so transitionary steps such as implementing plant-based defaults are relevant and necessary. This report therefore models the financial and environmental impacts of a range of animal-product reduction scenarios.

This report evaluates the impacts for universities of a transition to 100% plant-based catering. It will use financial and environmental modelling to estimate the cost per meal, CO2eq emissions, and water and land use. It will also model the financial and environmental impacts for universities (small, medium, and large) of a range of animal-product reduction scenarios leading towards 100% plant-based catering. Our findings demonstrate, in alignment with previous research and case studies, that transitioning to 100% plant-based catering enables university caterers to significantly reduce their environmental footprint without increasing catering costs.

2. Methods

Our full models are here, including findings, interactive elements, calculations, and graphs.

For simplicity, we predominantly used UK-specific data to build this model. For example, we used a UK wholesaler database to estimate food procurement costs, and used UK university data to estimate the financial and environmental impacts of various animal product reduction scenarios. The findings of this report are generally applicable internationally, however. Food procurement costs in the UK are similar to those in continental Europe, and the environmental data in this model is based on global averages rather than the UK specifically.

Some variables in the model are also adjustable, so that readers can adjust the model to suit their specific university context. For all these reasons, we have included EUR and USD conversions for all financial data.

2.1. Financial Model

To model the financial impact to caterers of transitioning to plant-based catering, we estimated the food costs of a typical university menu providing meat-based, vegetarian, and plant-based meals. To do this, we took 45 meals from a university sample menu

and estimated the cost per kilogram and portion size for each ingredient in each meal. This allowed us to calculate the cost per portion of each ingredient, and then the total food costs per meal. We then calculated the average cost of a meat-based, vegetarian, and plant-based meal. See Appendix C1 for our methods in full.

2.2. Environmental Model

To model the environmental impact of meat-based, vegetarian, and plant-based catering, we calculated the environmental impacts (CO2eq emissions, water use, land use) per kilogram of each ingredient using data from Poore & Nemecek (2018). This was then multiplied by our existing food portion sizes to calculate the environmental impacts of each ingredient per individual portion. This allowed us to calculate the environmental impacts (kgs CO2eq emitted, L of water used, m^2 of land used) for each meal. We then calculated the average environmental impacts of a meat-based, vegetarian, and plant-based meal. See Appendix C2 for our methods in full.

2.3. Transitionary Steps Model

To model the impacts of different levels of transition, we first calculated the impacts of a transition from a business-as-usual to a 100% plant-based scenario. We then modelled the impact of different levels of transition (30% or the implementation of plant-based default initiatives, 50%, and 70%) based on this total change, assuming that the marginal impacts of animal product consumption were consistent at different levels of consumption.

This model is not intended to perfectly predict all university caterers' food costs. University catering contexts are diverse and specific universities will have unique financial considerations. Instead, this model is intended to estimate financial and environmental savings in absolute terms for small, medium, and large universities. Some of the model's variables are also interactive to allow for readers to adjust the model's assumptions to fit their specific university context. See Appendix C3 for our methods in full.

3. Results

3.1. Food Costs

Average food costs for caterers were calculated based on three weeks of sample menus, featuring one meat-based, one vegetarian, and one plant-based meal each day, totaling 45 meals (15 of each meal type).

Meal costs ranged from £4.66 (€5.33/$6.11) per meal (for a beef lasagne with garlic cheese bread, fries, and a side salad) to £1.44 (€1.71/$1.88) per meal (for a plant-based jackfruit burger with pickled cabbage and potato wedges).

On average, the meat-based meals were most expensive, followed by the vegetarian meals, which were on average 11.52% cheaper. Plant-based meals were on average almost one-third cheaper than meat-based meals at 30.42%, and 21.37% cheaper than vegetarian meals. All 45 meals, their main ingredients, food costs, and environmental impacts, can be found in the appendices.

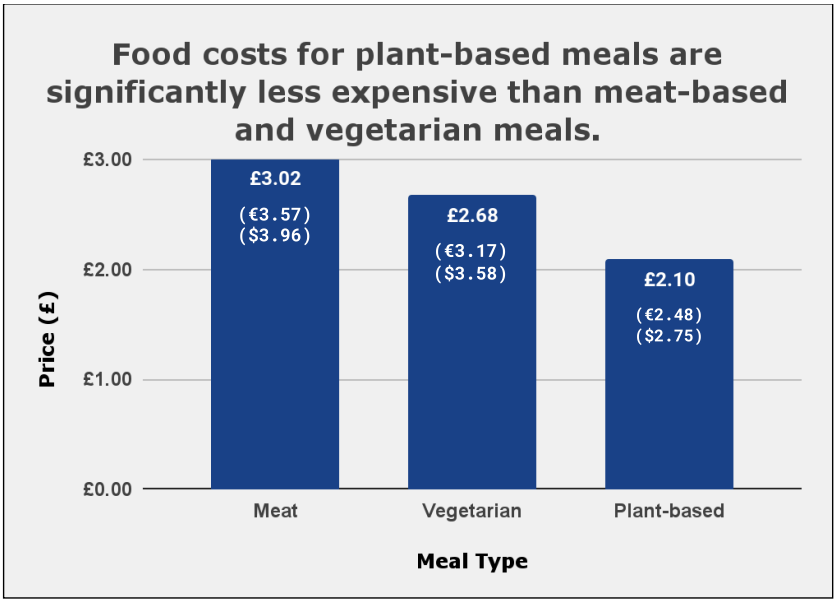

Figure 1: The average price of a meat-based, vegetarian, and plant-based meal.

3.2. Environmental Impacts

For the same three weeks of sample menus, average environmental impacts (CO2eq emissions, water use, land use) were calculated for a total of 45 meals (15 of each meal type). All environmental measures demonstrated a clear and consistent trend: the plant-based meals performed better compared to meat-based and vegetarian meals, often by a significant margin.

3.2.1. CO2eq Emissions

CO2eq emissions ranged from 11.4kg of CO2eq per meal (for a beef lasagne with garlic cheese bread, fries, and a side salad) to 0.5kg of CO2eq (for a chestnut and oyster mushroom kebab with pickled onions and chillies, and chips).

Meat-based meals had a significantly higher average carbon footprint (3.2kg of CO2eq per meal) compared to vegetarian (1.5kg per meal) and plant-based meals (0.5kg per meal).

Plant-based meals emit on average 84% less CO2eq compared to meat-based meals. Moreover, the average vegetarian meal still emits 3x as much CO2eq compared to the average plant-based meal. In no cases did the CO2eq emissions of a plant-based meal exceed that of a vegetarian or meat-based meal.

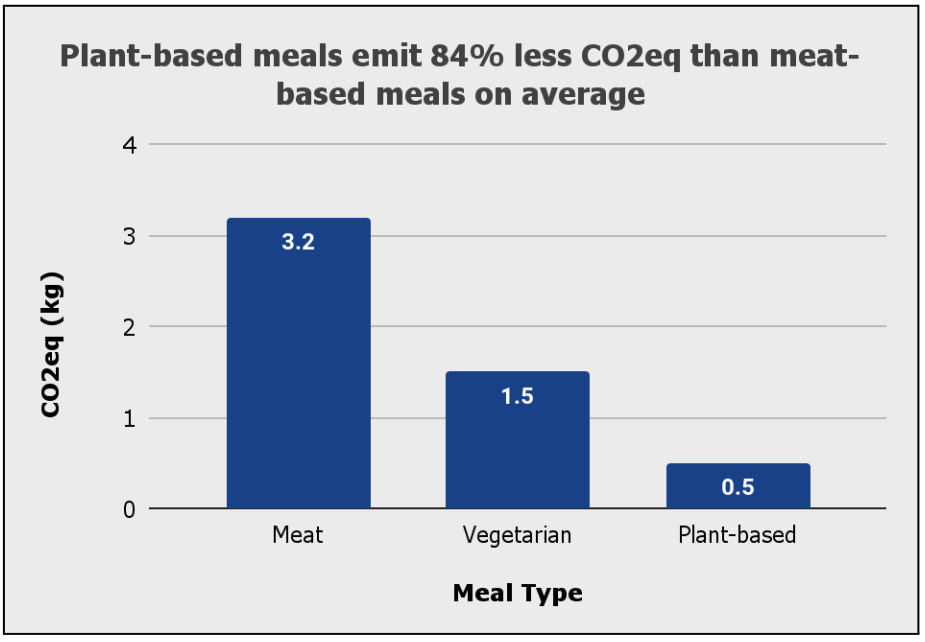

Figure 2: The average CO2eq emissions (kg) of a meat-based, vegetarian, and plant-based meal.

3.2.2. Water Use

Water use per meal ranged from 663.9 litres (for a meat feast pizza with mixed salad) to 66.5 litres (for a tofu and vegetable ramen with kimchi slaw).

Once again, meat-based meals had the highest average water use at 370.1 litres per meal. In contrast, the average plant-based meal used 69% less water, at 115.2 litres. Unlike other environmental measures, the average water use requirements for vegetarian meals (350.3 litres per meal) were just 5% lower than that of meat-based meals.

This is because dairy products, and cheese products in particular, demand particularly high volumes of water to produce. Hence, on average, only plant-based meals provide significant reductions in water use.

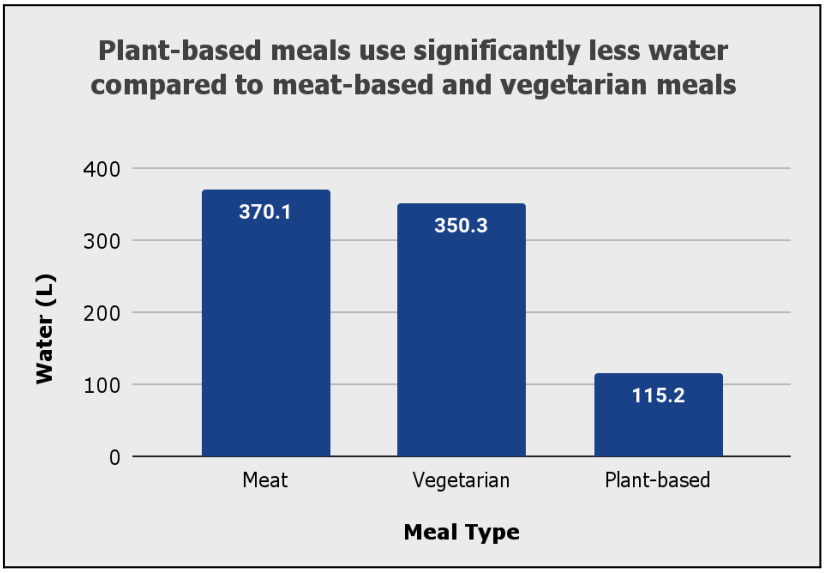

Figure 3: The average water use (litres) of a meat-based, vegetarian, and plant-based meal.

3.2.3. Land Use

Land use per meal ranged from 36m^2 per meal (for a beef lasagne with garlic cheese bread, fries, and a side salad) to 0.5m^2 (for a mushroom risotto with vegan cheese and a side salad).

Land use provides the greatest discrepancy in environmental impact between meat-based and plant-based meals. While the average amount of land required for a meat-based meal is 12.3m^2, the average for a plant-based meal is just 1.2m^2.

Meat-based meals therefore require on average over 10x the amount of land compared to plant-based meals. Plant-based meals also still require on average 73% less land than vegetarian meals.

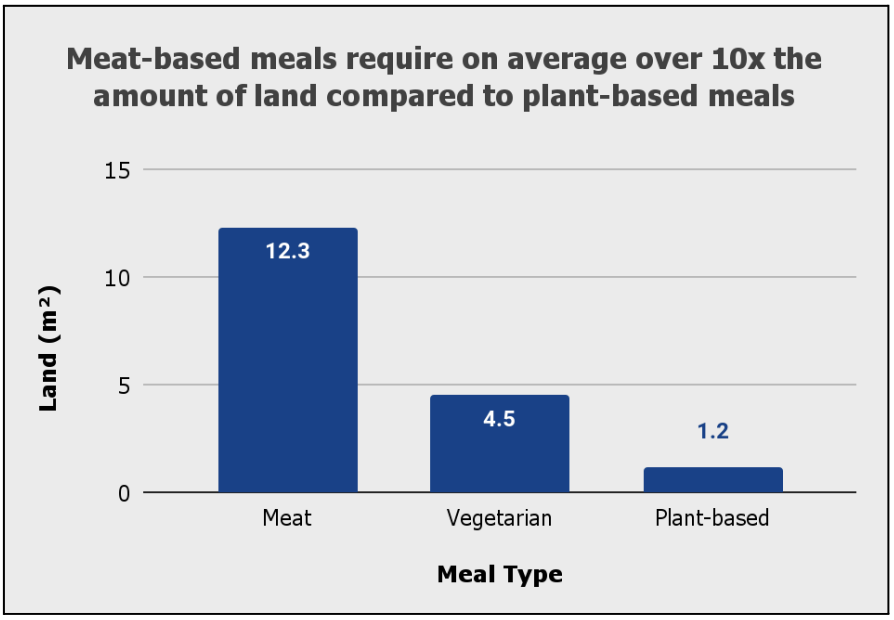

Figure 4: The average land use (m^2) of a meat-based, vegetarian, and plant-based meal.

3.3. Transitionary Steps

3.3.1. Food Cost and Environmental Impact Reductions Tables

The table below displays the total food costs, CO2eq emissions, water use, and land use for small (5,000 students), medium (10,000 students), and large (25,000 students) universities across five scenarios, progressing towards 100% plant-based catering. In all scenarios where animal product consumption is reduced and replaced with plant-based meals, food costs and all environmental impacts decrease.

		Scenario A (Business as usual)	Scenario B (Plant-based defaults/30% reduction)	Scenario C (50% reduction)	Scenario D (75% reduction)	Scenario E (100% reduction)
Small University (5,000 Students)	Food Costs (£)	£1,025,085	£941,261.10	£885,378.21	£815,524.60	£745,671.00
	CO2eq Emissions (kg)	919,552	701,505	556,140	374,435	192,729
	Water Use (L)	125,199,875	99,935,045	83,091,825	62,037,800	40,983,775
	Land Use (m2)	3,409,497	2,519,768	1,926,616	1,185,175	443,734
Medium University (10,000 Students)	Food Costs (£ per year)	£2,050,171	£1,882,522.19	£1,770,756.42	£1,631,049.21	£1,491,341.99
	CO2eq Emissions (kg per year)	1,839,103	1,403,010	1,112,281	748,870	385,459
	Water Use (L per year)	250,399,751	199,870,091	166,183,650	124,075,600	81,967,550
	Land Use (m2 per year)	6,818,995	5,039,537	3,853,231	2,370,350	887,468
Large University (25,000 Students)	Food Costs (£ per year)	£5,125,427	£4,706,305.48	£4,426,891.05	£4,077,623.02	£3,728,354.98
	CO2eq Emissions (kg per year)	4,597,758	3,507,525	2,780,702	1,872,175	963,647
	Water Use (L per year)	625,999,377	499,675,226	415,459,126	310,189,000	204,918,875
	Land Use (m2 per year)	17,047,487	12,598,842	9,633,079	5,925,874	2,218,670

Table 1: The expected food cost and environmental savings for small, medium, and large universities under incremental stages of a 100% plant-based catering transition. .

3.3.4. Food Cost and Environmental Impact Reductions Graphs

The graphs below visualise the reductions in food costs and environmental impacts for a medium university. Reductions are proportionally identical for small and large universities.

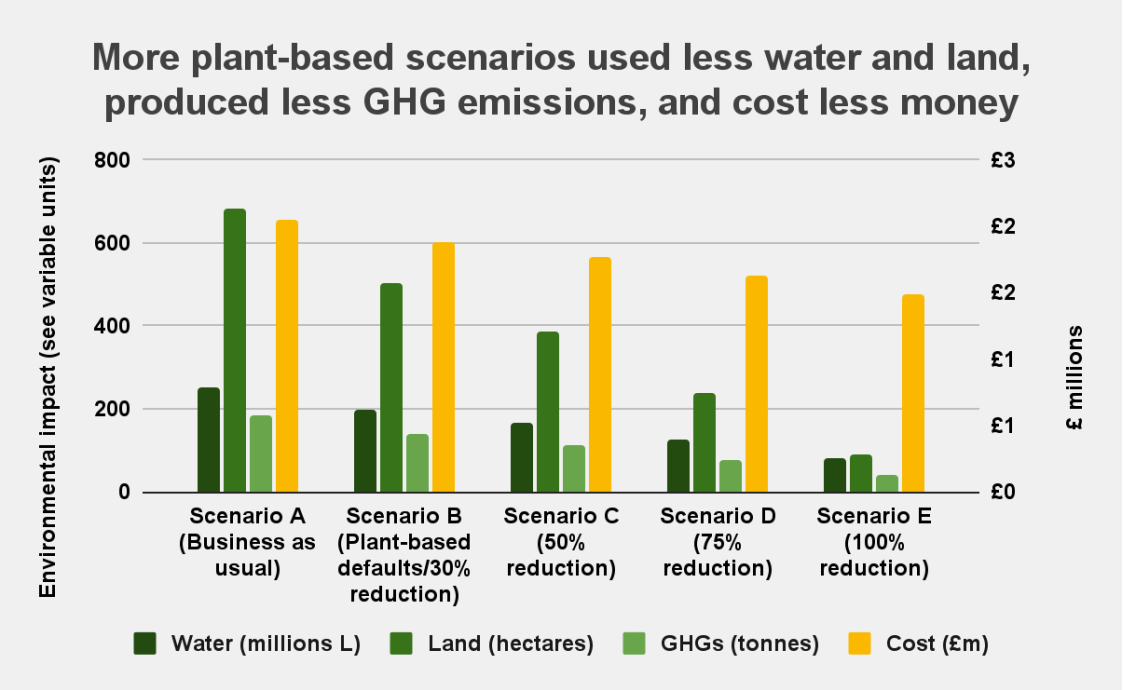

Figure 5: The financial and environmental impacts of university catering for a medium university at 5 scenarios towards 100% plant-based catering.

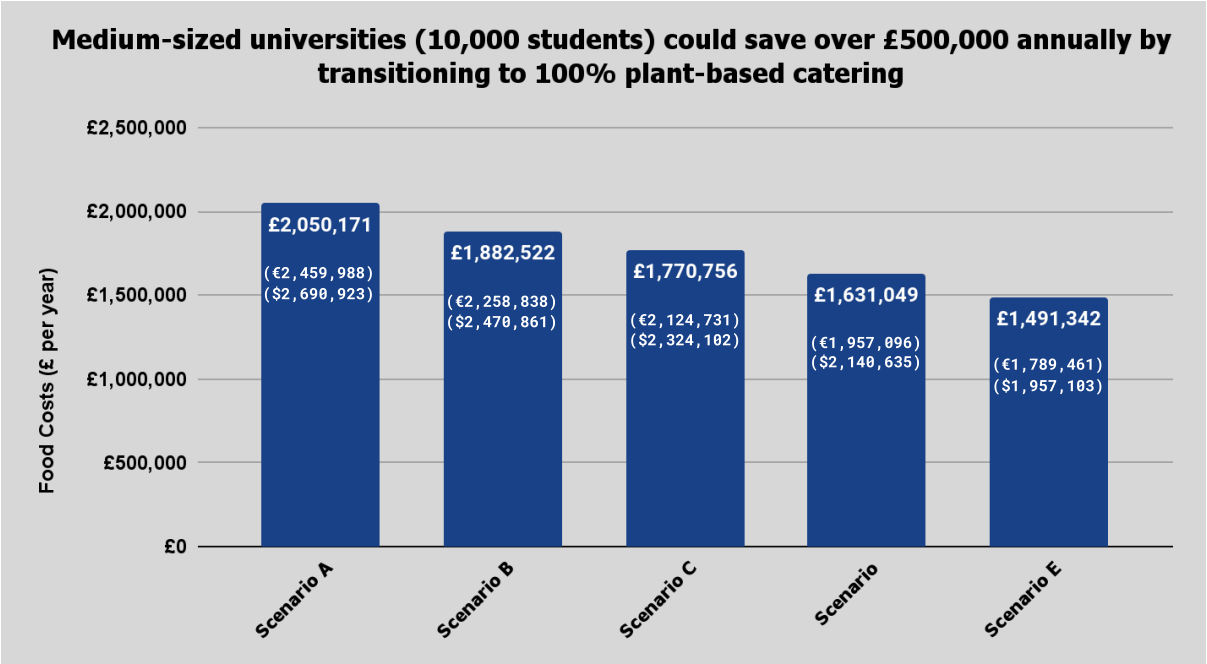

Figure 6: The expected food cost savings for a medium university under incremental stages of a 100% plant-based catering transition.

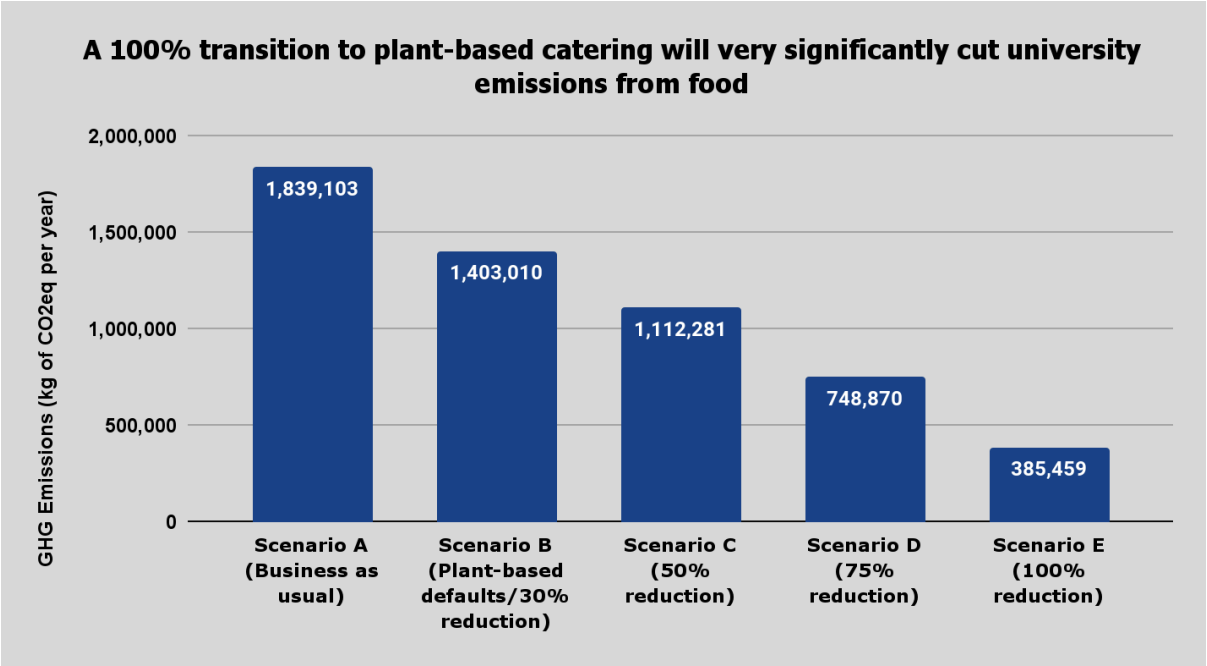

Figure 7: The expected CO2eq savings for a medium university under incremental stages of a 100% plant-based catering transition.

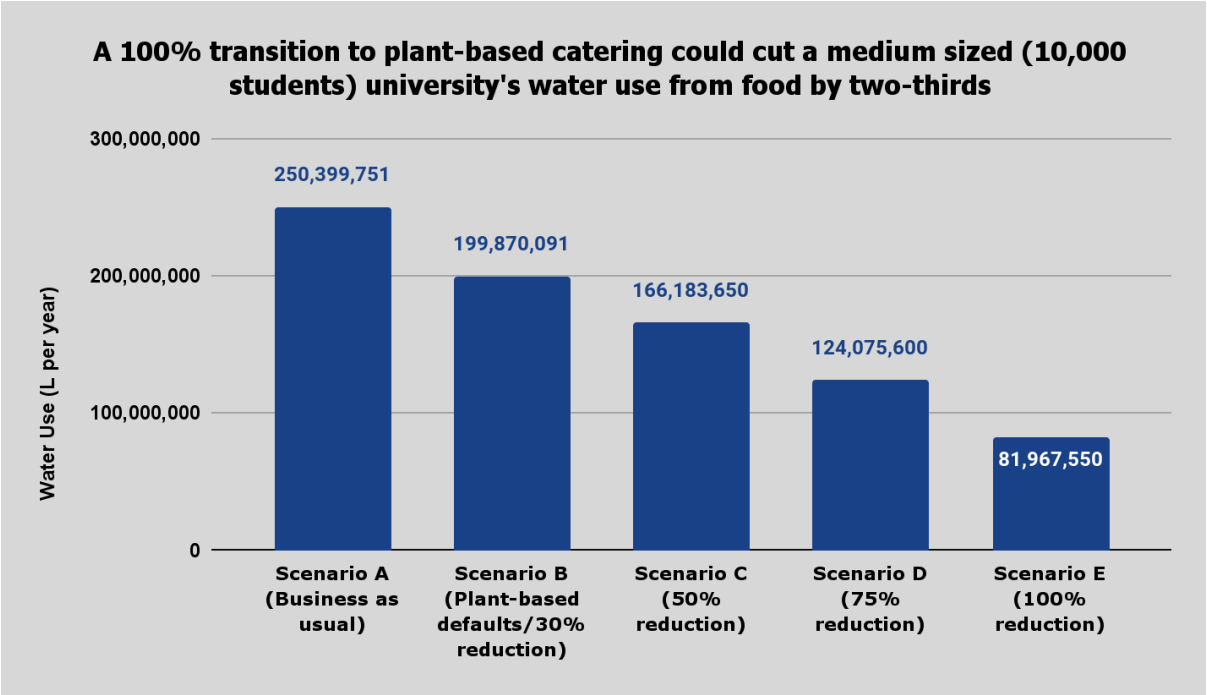

Figure 8: The expected water use savings for a medium university under incremental stages of a 100% plant-based catering transition.

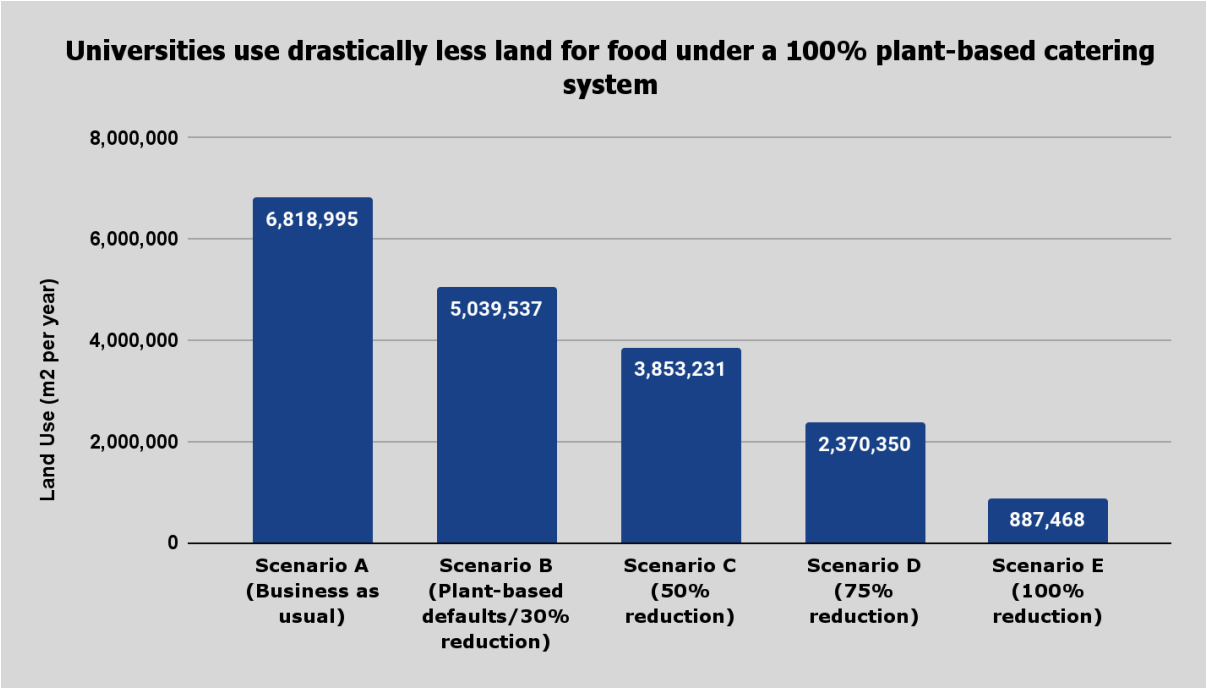

Figure 9: The expected land use savings for a medium university under incremental stages of a 100% plant-based catering transition.

4. Discussion

Taken together, the results of this model provide a compelling case for universities to transition to plant-based catering systems.

Firstly, these findings demonstrate, in line with a wealth of academic literature, that plant-based catering can significantly mitigate the environmental harms of food production and provision. This is consistent and substantial across a range of environmental measures.

It is worth noting the extent to which plant-based meals outperform meat-based and vegetarian meals on all environmental measures assessed here. Not only do plant-based meals emit a fraction of the emissions of meat-based meals (84% less on average) , but in this model, the emissions of a plant-based meal never exceeded that of a meat-based *or* vegetarian meal. For medium sized universities, a 100% plant-based transition could cut annual university food-related CO2eq emissions from 1.8 million kilograms to 385,459 kilograms.

Land-use savings were even more pronounced, with plant-based meals on average using almost 4x less land than vegetarian meals, and over 10x less land than meat based meals. For a university of 10,000 students, a 100% plant-based transition would slash land use from 6.8 million m^2 per year to 0.89 million m^2, an 87% reduction. Even simply implementing plant-based default initiatives would reduce land use by almost 2 million m^2.

Furthermore, plant-based meals required significantly less water use than both vegetarian and meat-based meals, which demanded almost identical volumes of water (370.1 litres for meat-based meals, and 350.3 litres for vegetarian meals, on average).

This data therefore shows that plant-based meals are not only the most environmentally friendly choice by a significant margin, they are also the only consistently environmentally friendly option across all environmental measures assessed here. As a result, the most effective and comprehensive step university caterers can take to significantly reduce their environmental footprint is to transition to a plant-based catering system.

Secondly, these findings also show that a plant-based catering transition is not only an environmental imperative but also a cost-effective measure.

While the financial implications of a plant-based transition are sometimes perceived as a barrier to implementation, this model found that the wholesale price of plant-based meal ingredients is on average 30% cheaper than meat-based meals, and 20% cheaper than vegetarian meals. Adopting plant-based catering can therefore reduce food procurement costs for universities.

Moreover, when combining menu changes with simple changes to choice architecture that encourage diners to choose plant-based options, caterers would be able to maintain or even increase their profit margins while transitioning. When applied to a medium university of 10,000 students, we found the potential for cost savings of over £500,000 (€600,000/$650,000) annually. Far from being a barrier to transition, conscientiously adopting and promoting plant-based menus promises significant savings in food procurement costs.

Cost reductions in food procurement could also be passed on to students, enabling student access to more affordable meals. In a cost of living crisis where over 40% of students are reported to be food insecure, this represents a significant beneficial spillover effect.[14] A plant-based transition is therefore not only environmentally superior and financially promising, it is also an opportunity to tackle student food security, and enable increased access to nutritious, sustainable meals.

In conclusion, this model demonstrates the many significant environmental benefits of transitioning to plant-based catering for universities. Importantly, this shift not only supports sustainability goals but also offers a cost-effective solution, as plant-based ingredients are on average significantly cheaper than meat-based or vegetarian options. The potential for cost savings could also improve student food security, if savings were passed onto students. A plant-based university transition therefore has significant environmental and social benefits, and can be achieved without imposing major added costs to caterers, institutions or customers.

[14] Food Standards Agency. (2023, January 10). Food behaviours in the UK student population: executive summary. Available at https://www.food.gov.uk/research/food-behaviours-in-the-uk-student-population-executive-summary

5. Policy Recommendations

1. Plan a Roadmap: Universities should publicly commit to and begin planning a phased transition to 100% plant-based catering on campus by 2030. Universities should set clear annual targets for increasing the proportion of plant-based meals. This is a crucial first step in signalling commitment to a just and sustainable food system, and in realising the financial and environmental benefits of a plant-based transition.

2. Engage and Educate Stakeholders: Universities should engage with, educate, and 'bring along' stakeholders, including Students' Union leaders and prospective and current students, as well as catering managers, commercial directors, chefs, directors of estates, and sustainability managers. All these parties should recognise the benefits of transitioning at the student, university, and societal level. Menu options should be diverse, nutritious and appealing as well as environmentally and financially advantageous.

3. Implement Defaults: The evidence is clear: defaults work. They can significantly increase student engagement with plant-based eating while being relatively straightforward to implement. These simple choice architecture changes can signal commitment to environmental sustainability, help to normalise plant-based eating, and begin to realise the benefits of a plant-based transition for universities.

4. Make use of support services: Various non-profit and other organisations exist to support caterers in transitioning to more plant-based menus. Humane Society International/UK is one such organisation that has experience addressing the barriers and opportunities faced by catering professionals when implementing plant-based menu items. Its Forward Food programme offers free support to caterers seeking to realise the benefits of a shift towards plant-centric menus. The programme has already collaborated with over 30 universities, providing tailored support to catering teams on the most effective ways to create and market tasty and nutritious plant-based options. This includes hands-on plant-based culinary trainings, evidence-backed menu formulation strategies, and bespoke food sustainability reports.

6. Appendices

Appendix A - Full list of sample meals and their estimated financial and environmental impacts

Group	Meal / Ingredient	Total cost	Total CO2e (kg)	Total water use (L)	Total land use (m2)
Meat	Lamb, mint, apricot koftas; wholemeal pitta; chilli yogurt, tabolleuh salad	£3.10	4.1	302.4	34.5
Meat	Salmon & Smoked Haddock Fishcake, Free Range Poached Egg, Shaved Fennel Salad	£3.77	1.4	317.5	1.2
Meat	Pork Belly, Soy Egg, Rice Noodle and Pak Choi Ramen	£2.94	1.8	377.6	2.3
Meat	Butter Chicken, Masala Sauce, Basmati Rice, Roti Bread, Onion & Coriander Bhaji	£2.17	1.8	286.1	2.6
Meat	Catch of the day, Chunky Chips, Garden or Mushy Peas, Homemade Tartare, Lemon	£2.84	1.7	408.1	2.2
Meat	Lamb Keema Curry, Naan Bread, Carrot Pickle, Rice, Mint & Coriander Chutney, Bombay Potatoes	£3.62	4.6	410.3	34.8
Meat	Meat Feast Pizza, Pepperoni, Spicy Beef, Mushroom, Green pepper & Grilled Chicken, Mixed Salad	£3.81	7.6	663.9	22.2
Meat	Crispy Chicken Katsu Curry, Steamed Chinese Cabbage & Pak Choi, Long Grain Rice,	£2.33	1.4	200.7	1.5
Meat	Cumberland Sausages, Caramelised Red Onion Gravy, Creamy Maris Piper Mashed Potatoes, Roasted Leeks	£2.84	1.5	180.5	1.9
Meat	Catch of the day, Chunky Chips, Garden or Mushy Peas, Homemade Tartare, Lemon	£2.84	1.7	408.1	2.2
Meat	Japanese Hirata buns with breaded chicken pickled cucumber salad cutty or sweet chilli dip	£2.80	1.3	157.3	2.0

Meat	Beef Lasagne served with red onion garlic cheese bread side salad fries	£4.66	11.4	663.5	36.0
Meat	Jollof rice, Mince lamb cape Mala curry, Ethiopian lentil dhal, Fresh salad	£2.38	4.2	318.3	33.8
Meat	Tortellini with spinach in a creamy mushroom chive sauce Marinated olives ciabatta bread	£2.42	1.8	448.9	5.6
Meat	Catch of the day, Chunky Chips, Garden or Mushy Peas, Homemade Tartare, Lemon	£2.84	1.7	408.1	2.2
Vegetarian	Grilled Halloumi; Sweet Potato; Homemade Flatbread; Toasted Corn Salsa	£2.65	2.8	658.0	9.4
Vegetarian	Mediterranean Vegetable, Spinach & Ricotta Lasagne, Toasted Garlic Ciabatta	£2.74	2.3	543.7	7.3
Vegetarian	Red Pepper, Feta Cheese, Roquette & Basil Frittata, Steamed Fine Beans	£2.54	1.3	246.8	3.3
Vegetarian	Roasted Root Vegetable, Goats Cheese & Thyme Wellington, Steamed Greens	£1.75	1.0	220.8	3.0
Vegetarian	Roasted Squash, Sage & Garden Pea Risotto, Shaved Parmesan, Green Salad	£2.08	1.2	344.3	3.7
Vegetarian	Tender stem Broccoli, Rice Noodle, Shitake Mushroom & Sesame Stir Fry, Crispy Tofu	£3.36	1.0	232.8	1.1
Vegetarian	Pulled Jackfruit & Sweet Potato Tacos, Pico de Gallo, Sour Cream, Black Beans, Burnt Corn Cob	£3.33	1.0	148.7	3.0
Vegetarian	Macaroni Cheese Burger, Sriracha Sauce, Cos Lettuce, Brioche Bun & French Fries	£1.90	1.1	303.4	3.5
Vegetarian	Cajun Spiced Vegetable, Black Bean & Cheddar Quesadilla, Tomato Salsa, Guacamole & Rice	£2.89	1.7	453.7	5.8
Vegetarian	Buffalo Mozzarella, Roquette, Tomato & Roasted Pepper Pizza, Mixed Green Salad	£2.85	1.8	421.5	5.8

Vegetarian	Roasted vegetable risotto topped with sundried tomato's fresh herbs cheese straws	£3.10	1.2	354.3	3.1
Vegetarian	Pulled Jackfruit & Sweet Potato Tacos, Pico de Gallo, Sour Cream, Black Beans, Burnt Corn Cob	£3.33	1.0	148.7	3.0
Vegetarian	Macaroni Cheese Burger, Sriracha Sauce, Cos Lettuce, Brioche Bun & French Fries	£1.90	1.1	303.4	3.5
Vegetarian	Cajun Spiced Vegetable, Black Bean & Cheddar Quesadilla, Tomato Salsa, Guacamole & Rice	£2.89	1.7	453.7	5.8
Vegetarian	Buffalo Mozzarella, Roquette, Tomato & Roasted Pepper Pizza, Mixed Green Salad	£2.85	1.8	421.5	5.8
Plant-based	Broccoli; Tofu & Aubergine Thai Yellow Curry; Steamed Rice; Raw Asian Slaw	£2.38	0.7	157.6	0.7
Plant-based	Cauliflower, Squash & Chickpea Tagine, Tabbouleh, Coconut Yoghurt	£1.99	0.8	150.3	2.4
Plant-based	Swarm Chestnut & Oyster Mushroom Kebab, Pickled Onion & Chillies & Chips	£2.43	0.3	76.4	0.6
Plant-based	Beetroot & Chickpea Burger, Toasted Bun, Vegan Mayo, Avocado & Sweet Potato Fries	£1.51	0.3	86.7	1.2
Plant-based	Chargrilled Pepper, Vegan Cheese, Black Olive & Baby Spinach Pizza, Green Salad	£1.94	0.5	143.2	1.1
Plant-based	Vegan Macaroni Cheese, Leek, Baby Spinach Bake, Dressed Baby Gem & Cherry Tomato Salad	£2.49	0.6	117.8	0.8
Plant-based	Rose Harissa Roasted Cauliflower Steak, Mint Tabbouleh, Pomegranate Coconut Yoghurt, Roasted Squash	£2.06	0.3	73.7	0.5
Plant-based	Quinoa Falafel Buda Bowl, Lemon & Chilli Dressing, Edamame, Pickled Cabbage, Crispy Tofu	£2.29	0.6	79.8	1.7
Plant-based	Spiced Jackfruit Burger, Vegan Bun Kale and avocado slaw, Lime Vegan Mayo, Sweet Potato Fries	£2.54	0.4	98.5	0.6

Plant-based	Pak Choi, Beansprout, Shitake Mushroom & Crispy Chili Tofu Ramen, Kimchi Slaw	£2.51	0.5	66.5	0.6
Plant-based	North African vegetable tagine cauliflower cous cous pita bread houmous	£2.31	0.9	220.1	3.9
Plant-based	Vegetable quinoa porridge with roasted squash topped with a sesame seed crumb	£1.69	0.4	68.2	1.0
Plant-based	Pulled BBQ Jackfruit in a vegan brioche bun pickled red cabbage sea salt wedges	£1.45	0.4	92.7	0.7
Plant-based	Portabella mushroom topped with barley risotto topped with vegan cheese side salad	£1.88	0.5	73.3	0.5
Plant-based	Mixed bean chipotle chili with roasted squash puree tomato rice	£1.94	0.8	222.7	2.6

Appendix B - Model specifications and assumptions

Appendix B1 – Financial model

To model the financial impact to caterers of transitioning to plant-based catering, we estimated the food costs of a typical university menu providing meat-based, vegetarian, and plant-based meals. University College London provides 4 weeks of sample menus that detail typical meals provided by its catering services. For 3 of those weeks, it provides a meat-based, vegetarian, and plant-based meal from Monday to Friday.

To estimate the food costs of each meal, individual ingredient prices (cost per kg of product) were estimated using an online compendium from Brakes, a leading UK wholesale food supplier. The portion sizes for each ingredient was estimated using the recommended guidelines from the British Dietetic Association. If portion size for an ingredient was unclear or specific to a particular meal, it was estimated using similar recipes from BBC Good Food.

For each ingredient, the cost per kilogram was multiplied by the portion size divided by 1000 to determine the cost per portion. The cost of each ingredient in a given meal was then added together to calculate the total food provision cost for the meal. This provided us with the total food costs for 3 weeks of typical university catering, encompassing 15 meat-based, 15 vegetarian, and 15 plant-based meals.

Appendix B2 – Environmental model

To model the environmental impact of meat-based, vegetarian, and plant-based catering, data from Poore & Nemecek (2018) was combined with our existing menu items and food portion estimates. Poore & Nemecek (2018) provided the largest metaanalysis of the environmental impact of foodstuffs to date, encompassing data from over 38,000 farms and 1,600 processors, packaging types, and retailers across 119 countries. It was chosen for its rigorous and comprehensive assessment of the multiple environmental impacts of 43 food groups.

CO2eq emissions, water use, and land use per kilogram of food were taken from Poore & Nemecek (2018) and multiplied by our existing food portion sizes to calculate the environmental impacts of each ingredient per individual portion. Since our model's ingredient list was more comprehensive than Poore & Nemecek's data, some generalisations were necessary. For example, dairy butter, yoghurt, and cream were all assigned the environmental impact data for milk, and dairy alternatives such as vegan

cheese were assigned the data for soy milk. Generalisations were always made conservatively. A full list of assumptions is available in appendix C.

As with the financial element of the model, the ultimate environmental impacts of a given meal were calculated by adding the environmental impacts per portion of each ingredient in a meal together. This provided us with the total environmental impacts (CO2eq emissions, water use, land use) for 3 weeks of typical university catering, encompassing 15 meat-based, 15 vegetarian, and 15 plant-based meals.

Appendix B3 – Transitionary steps model

Lastly, we modelled the financial and environmental impacts of a transition to 100% plant-based catering for small (5,000 students), medium (10,000 students), and large (25,000 students) universities. We established 5 scenarios on the path to 100% plant-based university catering.

Scenario A: Business as usual (no change in animal product consumption)
Scenario B: The implementation of plant-based defaults (or a 30% reduction in animal product consumption)
Scenario C: A 50% reduction in animal product consumption
Scenario D: A 75% reduction in animal product consumption
Scenario E: A 100% reduction in animal product consumption

To model the expected cost and environmental savings for any scenario, we first needed to estimate the current proportion of students that choose meat-based, vegetarian, or vegan meals when dining at their university. To do this, we took data from a UK Food Standards Agency report which found that 5% of UK university students are 'mainly vegan', 10% are vegetarian, 18% are flexitarian, and the rest are omnivores alongside a small percentage of pescetarians (7%)[15]. These dietary categories are somewhat unclear: we do not know to what extent 'flexitarians' or students who are 'mainly vegan' consume animal products on campus. For simplicity, and to ensure our model's findings remained conservative, we assumed that 'mainly vegan' students always chose the plant-based meal and never consumed animal products. We also assumed that flexitarians always chose the vegetarian meal, and never consumed meat products. Because meals containing fish are categorised as meat-based meals, we included pescetarians in the meat-based meal category. As such, this model's findings

[15] Food Standards Agency. (2023, January 6). Exploring food behaviours in the UK student population: Interim findings. Available at https://www.food.gov.uk/research/behaviour-and-perception/exploring-food-behaviours-in-the-uk-student-population-interim-findings?print=1

are likely to be conservative, since we are generally assuming that students eat fewer animal products than they do in reality.

To model the expected reduction in animal product consumption from implementing plant-based defaults, we referred to a recent report which reviewed the literature on plant-based default initiatives[16]. Plant-based default initiatives make changes to the choice architecture of a food service environment to ensure plant-based meal options are the default choice, with meat-based options requiring a more active and intentional choice from the consumer. Examples include making meat dishes less prominent on menus, placing plant-based meals first in buffet lines, or requiring consumers to opt-in for pre-planned catered events. The review estimated that, on average, plant-based default initiatives increase meat-free orders by 30.45 percentage points. Because we are interested in a reduction in animal-product orders rather than just meat orders, we are assuming that the effect size of a plant-based default nudge is the same as a vegetarian default nudge. This is likely a safe assumption, in particular because other reviews find significantly higher effect sizes in general for meat-free default initiatives (one found an average effect size of 55%).[17] Hence, we assumed that a university implementing a plant-based default initiative would see a 30% reduction in animal product consumption.

For each scenario where animal product consumption decreased, the decrease was split proportionally between the meat-based and vegetarian meals. We therefore assume that transitioning to plant-based catering involves substituting both meat-based dishes and vegetarian dishes proportionally with plant-based dishes.

Appendix B4 – General assumptions

We needed to estimate the number of meals an average student consumed on campus per year. A typical academic year in the UK is ~219 days, or 60% of a calendar year. To do this, we first assumed that 10% of students had a structured university meal plan, such as residence at a catered student halls, or a meal card. We based this assumption on a TUCO report from 2018 which found 5% of students live in catered halls[18]. No reliable data on the percentage of students with meal cards was available, so we

[16] Bryant Research. (2024, January). The £2 billion NHS windfall: Why meat reduction matters. Available at https://bryantresearch.co.uk/wp-content/uploads/2024/02/2billion-NHS-Windfall-CAWF.pdf
[17] Better Food Foundation. The Research on Nudges: DefaultVeg Works. Available at https://www.betterfoodfoundation.org/research-and-reports/research-on-nudges-defaultveg-works/
[18] TUCO. (2018). Student eating & drinking habits report. Available at https://www.tuco.ac.uk/sites/default/files/2019-07/Student-ED-Habits-final2018.pdf

assumed this amounted to a further 5%. We assumed that students on meal plans have on average 2 meals per day at university, and that students without have 0.5. We also assumed that on average Students' Unions own 50% of a university's food outlets. In reality, this percentage will vary widely from university to university.

It is important to note that we do not expect this model to be completely precise in its estimates of the financial and environmental costs of university catering. The model is focussed on the impacts of food procurement specifically, and so does not account for food waste, transport, or other fixed costs.

Some data, such as the percentage of university food outlets owned by the Students' Union, was not readily available. To ensure the model remains reliable, we have made this function, as well as the size of the university, an interactive feature of the model. Readers can therefore change the figures we have assumed to better fit the specific data relevant for their university. A link to our models is available in the Methods section of the report. Additionally, it is worth noting that even though food costs in absolute terms might vary from university to university, the proportional decrease in costs (based on the average cost of each meal type), is the same. This means that even if a specific university's annual food cost expenditure is different to the model's estimate, they can still expect food costs to reduce significantly as they transition to plant-based catering.

This allowed us to model the environmental and financial impacts of 5 scenarios leading towards a 100% plant-based university catering system. We calculated the total food costs, greenhouse gas emissions, water use, and land use for meat-based, vegetarian, and plant-based meals in different scenarios for small, medium, and large universities. First, we multiplied the average cost or environmental footprint of each meal type by the proportion of each meal type in a given scenario and the number of students at the university. This gave us the total cost or impact per mealtime. Finally, we multiplied this by the average number of meals a student eats per year—71—to estimate the total annual cost or impact.

Appendix C - Full assumptions from Poore & Nemecek (2018) Data

Ingredient (Model)	Ingredient (Poore & Nemecek)
Butter; Yoghurt; Sour Cream	Milk
Coconut Yoghurt/Milk; Vegan Mayo; Vegan Cheese	Soy Milk
Olives	Olive Oil
Mushrooms	Other Vegetables
Tahini	Nuts
Sauces (e.g. BBQ, Sweet Chilli)	Tomato

Appendix D - Limitations

This model makes some estimations and generalisations which may somewhat affect the accuracy of the results. For example, because environmental impact data for dairy products including butter, yoghurt, and cream, or dairy alternatives such as vegan cheese, were not available in Poore and Nemecek's dataset, data for milk or soy milk was used in their place. These generalisations are not expected to significantly impact the overall outcome of the analysis, which remains granular and rigorous. Furthermore, the results of this model align with extensive academic literature demonstrating the significant and myriad environmental benefits of plant-based diets.

It is sometimes claimed that Poore & Nemecek's meta-analysis is not representative of British farming's environmental impacts, since their data displays global averages, whereas the UK has more sustainable farming practices. Importantly, however, this claim does not refute this report's findings, or diminish the importance of a plant-based transition.

Firstly, the negative environmental impacts of animal products compared to plant-based products are generally so great that, even if UK farming practices are significantly more sustainable than global averages, this would still not negate the importance of a plant-based transition.

Secondly, the extent to which British farming is more sustainable than global averages is unclear, and in some cases, doubtful. For example, while it is true that a large proportion of UK beef is grass fed, most studies conclude that grass-fed beef has a

greater environmental impact in terms of land use and GHG emissions compared to animals fed grains and soy[19].

Therefore, this criticism does not negate the findings of this report, as the environmental impacts of animal agriculture in the UK still point to the significant benefits of a plant-based transition.

[19] Herrero, M., Henderson, B., Havlík, P., Thornton, P. K., Conant, R. T., Smith, P., ... & Stehfest, E. (2016). Greenhouse gas mitigation potentials in the livestock sector. *Nature Climate Change*, 6(5), 452-461.

Thank you

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